

NetSecure Project Roadmap

Project Goal

- Build a beginner-friendly cybersecurity tool
- Scan IP addresses and detect open ports
- Identify running services like SSH, HTTP, FTP
- Detect insecure or risky services
- Store scan history in a database
- Display everything on a visual dashboard

What This Project Will Achieve

- Learn networking fundamentals
- Understand ports and services
- Learn Python practically
- Build backend APIs using Flask
- Learn database integration
- Create dashboards and charts
- Understand basic cybersecurity concepts
- Create a strong portfolio project for internships

Section 0 — Environment Setup

- Install Python, VS Code, and Git
- Learn terminal basics
- Create project folder
- Setup GitHub repository

Section 1 — Networking + Port Scanner

- Learn IP addresses and ports
- Understand socket programming

- Build basic port scanner
- Detect open ports

Section 2 — Service Detection

- Learn common network services
- Map ports to services
- Detect HTTP, SSH, FTP, etc.

Section 3 — Risk Detection

- Learn insecure protocols
- Detect risky services
- Assign risk levels like High/Medium/Low

Section 4 — Flask Backend

- Learn Flask basics
- Create backend server
- Build routes and APIs

Section 5 — Dashboard UI

- Build frontend using HTML/CSS
- Create scan input form
- Display scan results in tables

Section 6 — Integration

- Connect frontend with backend
- Run scanner from dashboard
- Show live results

Section 7 — Database Integration

- Store scan history

- Save ports, services, and risks
- Use SQLite or MySQL

Section 8 — Charts & Visualization

- Use Chart.js
- Create pie charts and graphs
- Visualize risks and ports

Section 9 — Project Polish

- Improve UI
- Add README and screenshots
- Prepare for GitHub and resume

Final Outcome

- Working Network Vulnerability Scanner
- Security Dashboard
- Risk Detection System
- Database-backed Scan History
- GitHub Portfolio Project
- Resume-worthy Cybersecurity Project

Project Flow



